

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-a136cea162b351a4c.jsonl`  
- SHA-256: `66e55e3480e1c6925f98c318689a6a692c577d5cf6f94fd5528024a7d8790b1b`  
- Source modified: `2026-06-16T03:20:38+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:20:05.057Z)

Source Extractor

Research question: "Research question: **What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMM/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k,

edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6716 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6716 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 375 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible."

Fetch and extract key claims from this source:

****URL:**** <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11177965/>

****Title:**** Distributional bias compromises leave-one-out cross-validation (Rao et al.)

****Found via:**** Small-n evaluation rigor for video segmentation search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
 - be a concrete, checkable statement (not vague generalities)
 - include a direct quote from the source as support
 - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-16T03:20:07.035Z)

I'll fetch the source and extract the relevant claims.

3. user (2026-06-16T03:20:13.904Z)

REDIRECT DETECTED: The URL redirects to a different host.

Original URL: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11177965/>

Redirect URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11177965/>

Status: 301 Moved Permanently

To complete your request, I need to fetch content from the redirected URL. Please use WebFetch

again with these parameters:

- url: "https://pmc.ncbi.nlm.nih.gov/articles/PMC11177965/"
- prompt: "Extract the main thesis and findings of this paper about distributional bias and leave-one-out cross-validation (LOOCV). Specifically: (1) What is the core problem with leave-one-out cross-validation that the paper identifies? (2) What is "distributional bias" as defined here? (3) What concrete claims do they make about how LOOCV distorts evaluation, induces anticorrelation, or biases performance metrics? (4) Any proposed fixes or mitigations (e.g., stratification, alternative CV schemes)? (5) Quantitative results or magnitudes of the bias. (6) The publication date and venue/journal. Provide direct verbatim quotes for each key claim."

4. user (2026-06-16T03:20:24.005Z)

Analysis of "Distributional Bias Compromises Leave-One-Out Cross-Validation"

(1) Core Problem Identified

The paper identifies a fundamental flaw in LOOCV: it creates systematic negative correlation between training fold averages and corresponding test labels. The authors state that "this approach creates a negative correlation between the average label of each training fold and the label of its corresponding test instance."

(2) Definition of Distributional Bias

Distributional bias occurs because LOOCV's structure inherently produces this anticorrelation. When a high-value instance is held out for testing, the training set's average becomes artificially lower (and vice versa), creating a systematic mismatch. The paper describes how "predictions are aggregated across the entire dataset to calculate common performance metrics."

(3) Claims About LOOCV Distortion

The authors claim this bias "tends to negatively impact performance evaluation and hyperparameter optimization." They further state the effect "can lead to a bias against stronger regularization," meaning LOOCV may incorrectly penalize well-regularized models.

(4) Proposed Solutions

The paper proposes "a generalizable rebalanced cross-validation approach that corrects for distributional bias for both classification and regression" as their primary mitigation strategy.

(5) Quantitative Results

The abstract does not provide specific numerical magnitudes, only stating they "demonstrate that our approach improves cross-validation performance evaluation in synthetic simulations, across machine learning benchmarks, and in several published analyses."

(6) Publication Details

****Preprint date:**** June 3, 2024 (Version 2: March 24, 2025)
****Venue:**** ArXiv preprint; peer-reviewed version published in *Science Advances*
****Authors:**** George I Austin, Itsik Pe'er, Tal Korem

5. assistant (2026-06-16T03:20:27.768Z)

I have the content. Let me assess and structure the findings.

This is a primary research paper (Austin, Pe'er, Korem), published as an arXiv preprint and in *Science Advances*. The content is directly relevant to Dimension 4 of the research question (small-n evaluation rigor, LOOCV bias).

6. user (2026-06-16T03:20:32.135Z)

Structured output provided successfully

7. assistant (2026-06-16T03:20:38.728Z)

Extracted and returned via StructuredOutput.

Summary: This is a primary research paper (Austin, Pe'er, Korem; arXiv preprint June 2024, peer-reviewed in *Science Advances*) directly relevant to Dimension 4 (small-n evaluation rigor) of the research question. It documents a concrete, falsifiable failure mode of LOOCV ?

"distributional bias," an induced negative correlation between each training fold's mean label and the held-out test label ? that biases performance evaluation, hyperparameter selection, and regularization choices, with a proposed rebalanced-CV fix. This bears on the project's leave-one-video-out (LOVO) eval harness: when aggregating predictions across all held-out videos at $n \geq 16$, LOOCV/RL00CV-style aggregation can systematically distort the leaderboard, supporting the research question's stated concern about "leave-one-group-out vs RL00CV bias."

One caveat: the abstract-level fetch did not surface specific quantitative bias magnitudes, so the "improves evaluation" claim is rated supporting rather than central, and no numeric effect size could be verified.